Whoops! I just got flagged by the cops...

databases evolved to attain a state where the human element could be taken out of the situation. Now you can store all kinds of data and not worry about giving it structure. That comes later. With IoT gaining momentum, the sheer volume of data that'll be produced calls for even quicker analysis of the data and with that comes Machine Learning algorithms that can sift through massive data sets to ascertain patterns or anomalies.

Machine learning is being used across a multitude of application scenarios. If you have a smartphone, then the very reason that Google Now, Siri or Cortana can easily understand voice commands with an Indian accent is because of Machine Learning. Millions of us have tried out these services and the more it gets to hear, the better it picks up. If you're into writing blogs then you've obviously come across Google's Adwords which can show you which keywords are the best for your topic and then rank keywords based on search volume. The more you feed any machine learning system, the more it learns and the faster it becomes. It's only outlier cases that can throw off such systems and that too for a short while. So how does machine learning work? Here's a quick primer:

Machine Learning requires you to feed a set of learning algorithms voluminous amount of data. This data is broken down into three parts. First comes the 'training set' which allows the ML algorithms to easily figure out how to look for patterns, which parameters to prioritise (or give the highest weights). This data set would constitute roughly 60% of the entire data. Then comes the 'validation set' which helps the ML algorithms fine tune the parameters of various patterns. This forms about 20% of the entire data and lastly, you have the 'test set'. Given all the learning the system has done, you can figure out how accurate the system is to see if further training is required before opening the flood gates.

Predicting crime with Machine Learning

WHAT KIND OF DATA?

Since we were talking about predicting crime, you can easily assume that criminal case files would be one of the first things to be fed to the system. Manual methods that have been used by analysts involved looking at variables like names of schools which were prone to have higher gang members and then cross-referencing that data with slang words on Twitter to see if anything was afoot. A human would assign weights to indicate importance of certain terms. For example, in India the word "Ghoda" literally translates to "Horse" but the erstwhile gangsters would use "Ghoda" to reference a pistol. So when a PCA system comes across such terms which seem a little offbeat for normal conversation then it gets flagged. And if this flagged incident seems to be centered around a physical location that's known for violence then you're looking at a higher probability of a crime being committed. The local law enforcement authority of the area can then be notified at which point a human can investigate. The system depends on validation so it will have to be notified if its 'hunch' was indeed on point. This is how latent Dirichlet allocation (LDA) functions in a PCA system.

And the data fed to such PCA systems would include:

- **Biometrics** – To recognise individuals.
- **Body language patterns** – To identify abnormal behavior
- **Travel history** – To see if Hot Zones are developing based on density of criminals
- **Media consumption** – To see if flagged content is being viewed
- **Monetary asset levels** – Lower levels of wealth would indicate a higher propensity to commit crime if the person has a criminal history
- **Exposure to criminals** – To see if they're in bad company
- **Feeds** – To track movement of individuals